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How many seconds will it take for a car that is traveling at a constant rate of 45 miles per hour to travel a distance of 220 yards? (1 mile = 1,760 yards)

A. 8

B. 9

C. 10

D. 11

E. 12

Arithmetic Rate Problem; Measurement Conversion

First, convert 45 miles per hour to yards per second. We know that 1 hour is equal to 3,600 seconds giving us the fraction $\frac{45}{3{,}600}$. Divide both sides of the fraction by 3,600 to get the miles per second which equals $\frac{1}{80}$ miles in one second. We know that 1 mile is equal to 1,760 yards so the nest step is $1{,}760 \times \frac{1}{80}$. Therefore, it will take the car 1 second to travel 22 yards; hence it will take 10 seconds for the car to travel 220 yards.

Alternatively, let t be the number of seconds it will take for the car to travel 220 yards.

$$220 \text{ yd} = \left(45 \frac{\text{mi}}{\text{hr}}\right)(t \text{ sec}) \qquad \text{distance = rate × time}$$

$$(220 \cancel{\text{yd}})\left(\frac{1 \text{ mi}}{1{,}760 \text{ yd}}\right) = \left(45 \frac{\text{mi}}{\cancel{\text{hr}}}\right)\left(\frac{1 \cancel{\text{hr}}}{3{,}600 \cancel{\text{sec}}}\right)(t \cancel{\text{sec}}) \qquad \begin{array}{l}\text{1 mi = 1,760 yd}\\ \text{1 hr = 3,600 sec}\end{array}$$

$$\frac{220}{1{,}760} \text{ mi} = \left(\frac{45}{3{,}600} t\right) \text{mi} \qquad \text{cancel units}$$

$$\frac{2 \times 11 \times 10}{11 \times 16 \times 10} = \left(\frac{45}{45 \times 80}\right) t \qquad \text{factor}$$

$$\frac{1}{8} = \frac{t}{80} \qquad \text{cancel common factors}$$

$$t = 10 \qquad \text{multiply both sides by 80}$$

The correct answer is C.

•

A store's selling price of $2,240 for a certain computer would yield a profit of 40 percent of the store's cost for the computer. What selling price would yield a profit of 50 percent of the computer's cost?

A. $2,400

B. $2,464

C. $2,650

D. $2,732

E. $2,800

Algebra Applied Problems

Let $\$C$ be the store's cost for the computer. Then the store's profit from selling the computer would be $\$(2{,}240 - C)$, which is equal to 40 percent of the store's cost.

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$\$(2{,}240 - C)$, which is equal to 40 percent of the store's cost.

$$\begin{aligned} 2{,}240 - C &= 0.4C && \text{given} \\ 2{,}240 &= 1.4C && \text{add } C \text{ to both sides} \\ C &= 1{,}600 && \text{divide both sides by } 1.4 \end{aligned}$$

Thus, the store's cost for the computer is \$1,600. Therefore, to yield a profit of 50 percent of the computer's cost, the selling price would have to be 1.5(\$1,600) = \$2,400.

The correct answer is A.

•

If a certain coin is flipped, the probability that the coin will land heads up is $\frac{1}{2}$. If the coin is flipped 5 times, what is the probability that it will land heads up on the first 3 flips and not on the last 2 flips?

A. $\frac{3}{5}$

B. $\frac{1}{2}$

C. $\frac{1}{5}$

D. $\frac{1}{8}$

E. $\frac{1}{32}$

Arithmetic Probability

Because the 5 flips of the coin are independent of each other, the probability of the coin landing heads up on the first 3 flips and tails up (i.e., not heads up) on the last 2 flips is

$P(H) \times P(H) \times P(H) \times P(T) \times P(T)$,

where $P(H)$ and $P(T)$ are the probabilities, respectively, that a single flip of the coin lands heads up and tails up. Since $P(T) = 1 - P(H)$ and $P(H) = \frac{1}{2}$, it follows that $P(T) = \frac{1}{2}$.

Therefore, the requested probability is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{32}$.

The correct answer is E.

•

The operation $\otimes$ is defined for all nonzero numbers a and b by $a \otimes b = \frac{a}{b} - \frac{b}{a}$. If x and y are nonzero numbers, which of the following statements must be true?

I. $x \otimes xy = x(1 \otimes y)$

II. $x \otimes y = -(y \otimes x)$

III. $\frac{1}{x} \otimes \frac{1}{y} = y \otimes x$

A. I only

B. II only

C. III only

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C. III only

D. I and II

E. II and III

Algebra Simplifying Algebraic Expressions

I. CAN BE FALSE: Let $x = 2$ and $y = 3$. Then $\frac{1}{y} - y = -\frac{8}{3}$ and $x\left(\frac{1}{y} - y\right) = -\frac{16}{3}$, so $x \otimes xy = x(1 \otimes y)$ can be false.

II. MUST BE TRUE: $x \otimes y = \frac{x}{y} - \frac{y}{x} = -\left(\frac{y}{x} - \frac{x}{y}\right) = -(y \otimes x)$

III. MUST BE TRUE:

$$\frac{1}{x} \otimes \frac{1}{y} = \frac{\frac{1}{x}}{\frac{1}{y}} - \frac{\frac{1}{y}}{\frac{1}{x}} = \frac{y}{x} - \frac{x}{y} = y \otimes x$$

The correct answer is E.

•

If each term in the sum $a_1 + a_2 + \cdots + a_n$ is either 7 or 77 and the sum equals 350, which of the following could be equal to n?

A. 38

B. 39

C. 40

D. 41

E. 42

Arithmetic Operations with Integers

Since $a_1 + a_2 + \ldots + a_n = 350 = 7 \times 50$, if each term in the sum were a 7, then n would equal 50. But the only answer options are the consecutive integers from 38 to 42. So, suppose $a_1 = 77 = 7 \times 11$, while each of the other terms is a 7. Then $a_1 + a_2 + \ldots + a_n = 7 \times 50 = 7 \times (11 + 39) = (7 \times 11) + (7 \times 39)$. In that case, in addition to $a_1 = 77$, the sum has 39 other terms, each equal to 7: $a_2, \ldots, a_{40}$. Thus, $n = 1 + 39 = 40$.

The correct answer is C.

•

(1.00001)(0.99999) – (1.00002)(0.99998) =

A. 0

B. 10^{-10}

C. $3(10^{-10})$

D. 10^{-5}

E. $3(10^{-5})$

Arithmetic Equations

(1.00001)(0.99999) − (1.00002)(0.99998) = (1 + 0.00001)(0.99999) − (1 + 0.00002)(0.99998) = (0.99999 + 0.00069999) − (0.99998 + 0.0000199996) = 0.9999999999 − 0.9999999996 = 0.0000000003 = $3(10^{-10})$.

The correct answer is C.

•

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A certain rectangular photograph has an area of $6\frac{1}{2}$ square feet. What is the area of the photograph in square inches? (1 foot = 12 inches)

A. 36

B. 78

C. 432

D. 864

E. 936

Arithmetic Rational Numbers

Since 1 foot is 12 inches, 1 square foot is 12^2 = 144 square inches. So, the photograph's area of 6.5 square feet is 6.5 × 144 = 936 square inches.

The correct answer is E.

•

The toll for crossing a certain bridge is $0.75 each crossing. Drivers who frequently use the bridge may instead purchase a sticker each month for $13.00 and then pay only $0.30 each crossing during that month. If a particular driver will cross the bridge twice on each of x days next month and will not cross the bridge on any other day, what is the least value of x for which this driver can save money by using the sticker?

A. 14

B. 15

C. 16

D. 28

E. 29

Algebra First-Degree Equations

We need to find the least whole-number value of x for which $0.75(2x) > 13 + 0.3(2x)$; that is, the least whole number value for which $(1.5 - 0.6)x = 0.9x > 13$, and thus for which $x > \frac{130}{9} = 14\frac{4}{9}$.

This value is 15.

The correct answer is B.

Questions 152 to 207 — Difficulty: Hard

152. Two numbers differ by 2 and sum to S. Which of the following is the greater of the numbers in terms of S?

A. $\frac{S}{2} - 1$

B. $\frac{S}{2}$

C. $\frac{S}{2} + \frac{1}{2}$

D. $\frac{S}{2} + 1$

E. $\frac{S}{2} + 2$

Algebra First-Degree Equations

Let x represent the greater of the two numbers that differ by 2. Then, $x - 2$ represents the lesser of the two numbers. The two numbers sum to S, so $x + (x - 2) = S$. It follows that $2x - 2 = S$, or $2x = S + 2$, or

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$x = \frac{S}{2} + 1.$

The correct answer is D.

153. If m is an integer and $m = 10^{32} - 32$, what is the sum of the digits of m?

A. 257

B. 264

C. 275

D. 284

E. 292

Arithmetic Arithmetic Operations

When written in standard base 10 notation, 10^{32} is the digit 1 followed by 32 digits of 0. Now consider the following subtractions and the digit pattern they suggest, a pattern which is easily seen to continue.

$$100 - 32 = 68$$

$$1{,}000 - 32 = 968$$

$$10{,}000 - 32 = 9{,}968$$

$$100{,}000 - 32 = 99{,}968$$

$$1{,}000{,}000 - 32 = 999{,}968$$

Using this digit pattern, $m = 10^{32} - 32$ in standard base 10 notation consists of 32 − 2 = 30 occurrences of the digit 9 followed by the digits 6 and 8. Therefore, the sum of the digits of m is 30(9) + 6 + 8 = 270 + 14 = 284.

The correct answer is D.

154. In a numerical table with 10 rows and 10 columns, each entry is either a 9 or a 10. If the number of 9s in the nth row is $n - 1$ for each n from 1 to 10, what is the average (arithmetic mean) of all the numbers in the table?

A. 9.45

B. 9.50

C. 9.55

D. 9.65

E. 9.70

Arithmetic Operations with Integers

There are (10)(10) = 100 entries in the table. In rows 1, 2, 3, …, 10, the number of 9s is 0, 1, 2, …, 9, respectively, giving a total of 0 + 1 + 2 + … + 9 = 45 entries with a 9. This leaves a total of 100 – 45 = 55 entries with a 10. Therefore, the sum of the 100 entries is 45(9) + 55(10) = 405 + 550 = 955, and the average of the 100 entries is $\frac{955}{100} = 9.55$

The correct answer is C.

155. In 2004, the cost of 1 year-long print subscription to a certain newspaper was $4 per week. In 2005, the newspaper introduced a new rate plan for 1 year-long print subscription: $3 per week for the first 40 weeks of 2005 and $2 per week for the remaining weeks of 2005. How much less did 1 year-long print subscription to this newspaper cost in 2005 than in 2004?

A. $64

B. $78

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C. $112

D. $144

E. $304

Arithmetic Applied Problems

The cost, in dollars, of 1 year-long print subscription in 2004 was 52(4) = 208 and the cost, in dollars, of 1 year-long print subscription in 2005 was 40(3) + 12(2) = 120 + 24 = 144. Therefore, the cost in 2005 was less than the cost in 2004 by 208 − 144 = 64 dollars.

The correct answer is A.

156. A positive integer n is a perfect number provided that the sum of all the positive factors of n, including 1 and n, is equal to $2n$. What is the sum of the reciprocals of all the positive factors of the perfect number 28?

A. $\frac{1}{4}$

B. $\frac{56}{27}$

C. 2

D. 3

E. 4

Arithmetic Properties of Integers

The factors of 28 are 1, 2, 4, 7, 14, and 28. Therefore, the sum of the reciprocals of the factors of 28 is

$$\frac{1}{1}+\frac{1}{2}+\frac{1}{4}+\frac{1}{7}+\frac{1}{14}+\frac{1}{28}=\frac{28}{28}+\frac{14}{28}+\frac{7}{28}+\frac{4}{28}+\frac{2}{28}+\frac{1}{28}=\frac{28+14+7+4+2+1}{28}=\frac{56}{28}=2.$$

The correct answer is C.

157. The infinite sequence $a_1, a_2, \cdots, a_n, \cdots$ is such that $a_1 = 2$, $a_2 = -3$, $a_3 = 5$, $a_4 = -1$, and $a_n = a_{n-4}$ for $n > 4$. What is the sum of the first 97 terms of the sequence?

A. 72

B. 74

C. 75

D. 78

E. 80

Arithmetic Sequences and Series

Because $a_n = a_{n-4}$ for $n > 4$, it follows that the terms of the sequence repeat in groups of 4 terms:

Values for n	Values for a_n
1, 2, 3, 4	2, –3, 5, –1
5, 6, 7, 8	2, –3, 5, –1
9, 10, 11, 12	2, –3, 5, –1
13, 14, 15, 16	2, –3, 5, –1

Thus, since 97 = 24(4) + 1, the sum of the first 97 terms can be grouped into 24 groups of 4 terms each, with one remaining term, which allows the sum to be easily found:

$$(a_1 + a_2 + a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \ldots +$$

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$(a_{93} + a_{94} + a_{95} + a_{96}) + a_{97}$

$= (2 - 3 + 5 - 1) + (2 - 3 + 5 - 1) + \ldots +$
$(2 - 3 + 5 - 1) + 2$

$= 24(2 - 3 + 5 - 1) + 2 = 24(3) + 2 = 74$

The correct answer is B.

158. The sequence $a_1, a_2, \ldots, a_n, \ldots$ is such that $a_n = 2a_{n-1} - x$ for all positive integers $n \geq 2$ and for a certain number x. If $a_5 = 99$ and $a_3 = 27$, what is the value of x?

A. 3

B. 9

C. 18

D. 36

E. 45

Algebra Sequences and Series

An expression for a_5 that involves x can be obtained using $a_3 = 27$ and applying the equation $a_n = 2a_{n-1} - x$ twice, once for $n = 4$ and once for $n = 5$.

$a_4 = 2a_3 - x$	using $a_n = 2a_{n-1} - x$ for $n = 4$
$= 2(27) - x$	using $a_3 = 27$
$a_5 = 2a_4 - x$	using $a_n = 2a_{n-1} - x$ for $n = 5$
$= 2[2(27) - x] - x$	using $a_4 = 2(27) - x$
$= 4(27) - 3x$	combine like terms

Therefore, using $a_5 = 99$, we have

$99 = 4(27) - 3x$	given
$3x = 4(27) - 99$	adding $(3x - 99)$ to both sides
$x = 4(9) - 33$	dividing both sides by 3
$x = 3$	arithmetic

The correct answer is A.

159. In a certain medical survey, 45 percent of the people surveyed had the type A antigen in their blood and 3 percent had both the type A antigen and the type B antigen. Which of the following is closest to the percent of those with the type A antigen who also had the type B antigen?

A. 1.35%

B. 6.67%

C. 13.50%

D. 15.00%

E. 42.00%

Arithmetic Applied Problems: Percents

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Arithmetic Applied Problems; Percents

Let n be the total number of people surveyed. Then, the proportion of the people who had type A who also had type B is $\frac{(3\%)\cancel{n}}{(45\%)\cancel{n}} = \frac{3}{45} = \frac{1}{15}$, which as a percent is approximately 6.67%. Note that by using $\frac{1}{15} = \frac{1}{3} \times \frac{1}{5}$, which equals $\frac{1}{3}$ of 20%, we can avoid dividing by a 2-digit integer.

The correct answer is B.

160. On a certain transatlantic crossing, 20 percent of a ship's passengers held round-trip tickets and also took their cars aboard the ship. If 60 percent of the passengers with round-trip tickets did not take their cars aboard the ship, what percent of the ship's passengers held round-trip tickets?

A. $33\frac{1}{3}\%$

B. 40%

C. 50%

D. 60%

E. $66\frac{2}{3}\%$

Arithmetic Percents

Since the number of passengers on the ship is immaterial, let the number of passengers on the ship be 100 for convenience. Let x be the number of passengers that held round-trip tickets. Then, since 20 percent of the passengers held a round-trip ticket and took their cars aboard the ship, 0.20(100) = 20 passengers held round-trip tickets and took their cars aboard the ship. The remaining passengers with round-trip tickets did not take their cars aboard, and they represent $0.6x$ (that is, 60 percent of the passengers with round-trip tickets). Thus $0.6x + 20 = x$, from which it follows that $20 = 0.4x$, and so $x = 50$. The percent of passengers with round-trip tickets is, then, $\frac{50}{100} = 50\%$.

The correct answer is C.

161. If x and k are integers and $(12^x)(4^{2x+1}) = (2^k)(3^2)$, what is the value of k?

A. 5

B. 7

C. 10

D. 12

E. 14

Arithmetic Exponents

Rewrite the expression on the left so that it is a product of powers of 2 and 3.

$$\begin{aligned}(12^x)(4^{2x+1}) &= \left[(3\cdot 2^2)^x\right]\left[(2^2)^{2x+1}\right] \\ &= \left(3^x\right)\left[(2^2)^x\right]\left[2^{2(2x+1)}\right] \\ &= (3^x)(2^{2x})(2^{4x+2}) \\ &= (3^x)(2^{6x+2})\end{aligned}$$

Then, since $(12^x)(4^{2x+1}) = (2^k)(3^2)$, it follows that $(3^x)(2^{6x+2}) = (2^k)(3^2) = (3^2)(2^k)$, so $x = 2$ and $k = 6x + 2$. Substituting 2 for x gives $k = 6(2) + 2 = 14$.

The correct answer is E.

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162. If S is the sum of the reciprocals of the 10 consecutive integers from 21 to 30, then S is between which of the following two fractions?

A. $\frac{1}{3}$ and $\frac{1}{2}$

B. $\frac{1}{4}$ and $\frac{1}{3}$

C. $\frac{1}{5}$ and $\frac{1}{4}$

D. $\frac{1}{6}$ and $\frac{1}{5}$

E. $\frac{1}{7}$ and $\frac{1}{6}$

Arithmetic Estimation

The value of $\frac{1}{21}+\frac{1}{22}+\frac{1}{23}+\ldots+\frac{1}{30}$ is LESS than $\frac{1}{20}+\frac{1}{20}+\frac{1}{20}+\ldots+\frac{1}{20}$ (10 numbers added), which equals $10\left(\frac{1}{20}\right)=\frac{1}{2}$, and GREATER than $\frac{1}{30}+\frac{1}{30}+\frac{1}{30}+\ldots+\frac{1}{30}$ (10 numbers added), which equals $10\left(\frac{1}{30}\right)=\frac{1}{3}$.

Therefore, the value of $\frac{1}{21}+\frac{1}{22}+\frac{1}{23}+\ldots+\frac{1}{30}$ is between $\frac{1}{3}$ and $\frac{1}{2}$.

The correct answer is A.

163. For every even positive integer m, $f(m)$ represents the product of all even integers from 2 to m, inclusive. For example, $f(12)=2\times4\times6\times8\times10\times12$. What is the greatest prime factor of $f(24)$?

A. 23

B. 19

C. 17

D. 13

E. 11

Arithmetic Properties of Integers

Rewriting $f(24)=2\times4\times6\times8\times10\times12\times14\times\ldots\times20\times22\times24$ as 2 × 4 × 2(3) × 8 × 2(5) × 12 × 2(7) × … × 20 × 2(11) × 24 shows that all of the prime numbers from 2 through 11 are factors of $f(24)$. The next prime number is 13, but 13 is not a factor of $f(24)$ because none of the even integers from 2 through 24 has 13 as a factor. Therefore, the largest prime factor of $f(24)$ is 11.

The correct answer is E.

$$3,\ k,\ 2,\ 8,\ m,\ 3$$

164. The arithmetic mean of the list of numbers above is 4. If k and m are integers and $k\neq m$ what is the median of the list?

A. 2

B. 2.5

C. 3

D. 3.5

E. 4

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Arithmetic Statistics

Since the $\text{arithmetic mean} = \dfrac{\text{sum of values}}{\text{number of values}}$, then $\dfrac{3+k+2+8+m+3}{6} = 4$, and so $\dfrac{16+k+m}{6} = 4$, $16+k+m = 24$, $k+m = 8$. Since $k \neq m$, then either $k < 4$ and $m > 4$ or $k > 4$ and $m < 4$. Because k and m are integers, either $k \leq 3$ and $m \geq 5$ or $k \geq 5$ and $m \leq 3$.

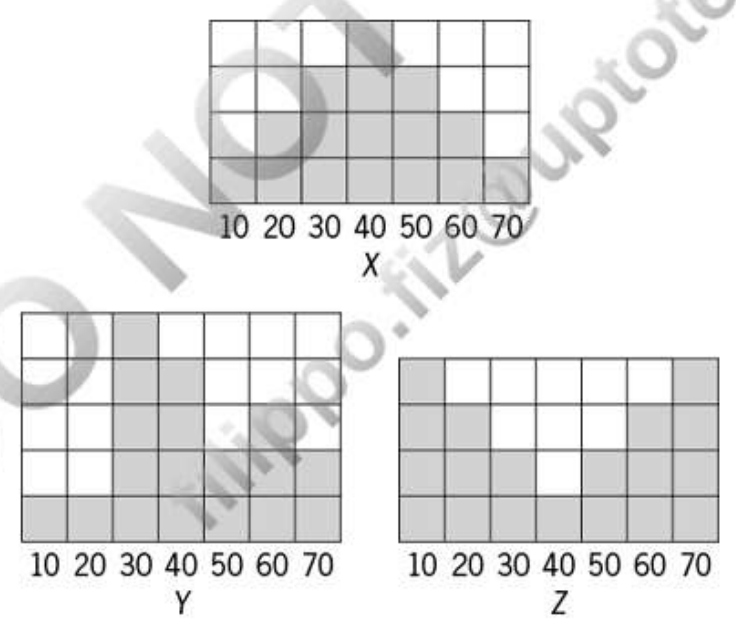
Case (i): If $k \leq 2$, then $m \geq 6$ and the six integers in ascending order are k, 2, 3, 3, m, 8 or k, 2, 3, 3, 8, m. The two middle integers are both 3 so the median is $\dfrac{3+3}{2} = 3$.

Case (ii): If $k = 3$, then $m = 5$ and the six integers in ascending order are 2, k, 3, 3, m, 8. The two middle integers are both 3 so the median is $\dfrac{3+3}{2} = 3$.

Case (iii): If $k = 5$, then $m = 3$ and the six integers in ascending order are 2, m, 3, 3, k, 8. The two middle integers are both 3 so the median is $\dfrac{3+3}{2} = 3$.

Case (iv): If $k \geq 6$, then $m \leq 2$ and the six integers in ascending order are m, 2, 3, 3, k, 8 or m, 2, 3, 3, 8, k. The two middle integers are both 3 so the median is $\dfrac{3+3}{2} = 3$.

The correct answer is C.



165. If the variables X, Y, and Z take on only the values 10, 20, 30, 40, 50, 60, or 70 with frequencies indicated by the shaded regions above, for which of the frequency distributions is the mean equal to the median?

A. X only

B. Y only

C. Z only

D. X and Y

E. X and Z

Arithmetic Statistics

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Arithmetic Statistics

The frequency distributions for both X and Z are symmetric about 40, and thus both X and Z have mean = median = 40. Therefore, any answer choice that does not include both X and Z can be eliminated. This leaves only answer choice E.

The correct answer is E.

$$2x + y = 12$$
$$|y| \leq 12$$

166. For how many ordered pairs (x, y) that are solutions of the system above are x and y both integers?

A. 7

B. 10

C. 12

D. 13

E. 14

Algebra Absolute Value

From $|y| \leq 12$, if y must be an integer, then y must be in the set $S = \{\pm 12, \pm 11, \pm 10, \ldots, \pm 3, \pm 2, \pm 1, 0\}$.

Since $2x + y = 12$, then $x = \dfrac{12 - y}{2}$. If x must be an integer, then $12 - y$ must be divisible by 2; that is, $12 - y$ must be even. Since 12 is even, $12 - y$ is even if and only if y is even. This eliminates all odd integers from S, leaving only the even integers $\pm 12, \pm 10, \pm 8, \pm 6, \pm 4, \pm 2$, and 0. Thus, there are 13 possible integer y-values, each with a corresponding integer x-value and, therefore, there are 13 ordered pairs (x, y), where x and y are both integers, that solve the system.

The correct answer is D.

167. The United States Mint produces coins in 1-cent, 5-cent, 10-cent, 25-cent, and 50-cent denominations. If a jar contains exactly 100 cents worth of these coins, which of the following could be the total number of coins in the jar?

I. 91

II. 81

III. 76

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

Arithmetic Operations with Integers

Letting p, n, d, q, and h, respectively, represent the numbers of pennies (1-cent coins), nickels (5-cent coins), dimes (10-cent coins), quarters (25-cent coins), and half-dollars (50-cent coins) with a total worth of 100 cents, it follows that $p + 5n + 10d + 25q + 50h = 100$. Then $p = 100 - (5n + 10d + 25q + 50h) = 5(20 - n - 2d - 5q - 10h)$, so p must be a multiple of 5.

I. If the jar contained 90 pennies and 1 dime, the total number of coins would be 90 + 1 = 91 and the coins would be worth 90 + 10 = 100 cents.

II. For the jar to contain 81 coins with a total worth of 100 cents, there could be 80 pennies, but then the one coin remaining would have to amount to 20 cents to make the coins' total worth 100 cents. This is not possible since none of the coins is a 20-cent coin. If there were 75 pennies, then the remaining 6 coins would have to amount to 25 cents. This is not possible because 6 coins of the next smallest denomination

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would have to amount to 25 cents. This is not possible because 6 coins of the next smallest denomination would be worth 30 cents. If there were 70 pennies, then the remaining 11 coins would have to amount to 30 cents. This is not possible because 11 coins of the next smallest denomination would be worth 55 cents. Continuing in this manner shows that it is not possible for the jar to contain 81 coins with a total worth of 100 cents.

III. If the jar contained 70 pennies and 6 nickels, the total number of coins would be 70 + 6 = 76 and the coins would be worth 70 + 30 = 100 cents.

The correct answer is D.

168. A certain university will select 1 of 7 candidates eligible to fill a position in the mathematics department and 2 of 10 candidates eligible to fill 2 identical positions in the computer science department. If none of the candidates is eligible for a position in both departments, how many different sets of 3 candidates are there to fill the 3 positions?

A. 42

B. 70

C. 140

D. 165

E. 315

Arithmetic Elementary Combinatorics

To fill the position in the math department, 1 candidate will be selected from a group of 7 eligible candidates, and so there are 7 sets of 1 candidate each to fill the position in the math department. To fill the positions in the computer science department, any one of the 10 eligible candidates can be chosen for the first position and any of the remaining 9 eligible candidates can be chosen for the second position, making a total of 10 × 9 = 90 sets of 2 candidates to fill the computer science positions. But this number includes the set in which Candidate A was chosen to fill the first position and Candidate B was chosen to fill the second position as well as the set in which Candidate B was chosen for the first position and Candidate A was chosen for the second position. These sets are not different essentially since the positions are identical and in both sets Candidates A and B are chosen to fill the 2 positions. Therefore, there are $\frac{90}{2} = 45$ sets of 2 candidates to fill the computer science positions. Then, using the multiplication principle, there are 7 × 45 = 315 different sets of 3 candidates to fill the 3 positions.

The correct answer is E.

169. A survey of employers found that during 1993 employment costs rose 3.5 percent, where employment costs consist of salary costs and fringe-benefit costs. If salary costs rose 3 percent and fringe-benefit costs rose 5.5 percent during 1993, then fringe-benefit costs represented what percent of employment costs at the beginning of 1993?

A. 16.5%

B. 20%

C. 35%

D. 55%

E. 65%

Algebra; Arithmetic First-Degree Equations; Percents

Let E represent employment costs, S represent salary costs, and F represent fringe-benefit costs. Then $E = S + F$. An increase of 3 percent in salary costs and a 5.5 percent increase in fringe-benefit costs resulted in a 3.5 percent increase in employment costs. Therefore $1.03S + 1.055F = 1.035E$. But, $E = S + F$, so $1.03S + 1.055F = 1.035(S + F) = 1.035S + 1.035F$.

Combining like terms gives $(1.055 - 1.035)F = (1.035 - 1.03)S$ or $0.02F = 0.005S$. Then, $S = \frac{0.02}{0.005}F = 4F$. Thus, since $E = S + F$, it follows that $E = 4F + F = 5F$. Then, F as a percent of E is $\frac{F}{E} = \frac{F}{5F} = \frac{1}{5} = 20\%$.

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E 5F 5

The correct answer is B.

170. The subsets of the set $\{w, x, y\}$ are $\{w\}$, $\{x\}$, $\{y\}$, $\{w, x\}$, $\{w, y\}$, $\{x, y\}$, $\{w, x, y\}$, and $\{\ \}$ (the empty subset). How many subsets of the set $\{w, x, y, z\}$ contain w?

A. Four

B. Five

C. Seven

D. Eight

E. Sixteen

Arithmetic Sets

As shown in the table, the subsets of $\{w, x, y, z\}$ can be organized into two columns, those subsets of $\{w, x, y, z\}$ that do not contain w (left column) and the corresponding subsets of $\{w, x, y, z\}$ that contain w (right column), and each of these collections has the same number of sets. Therefore, there are 8 subsets of $\{w, x, y, z\}$ that contain w.

subsets not containing w	subsets containing w
$\{\ \}$	$\{w\}$
$\{x\}$	$\{w, x\}$
$\{y\}$	$\{w, y\}$
$\{z\}$	$\{w, z\}$
$\{x, y\}$	$\{w, x, y\}$
$\{x, z\}$	$\{w, x, z\}$
$\{y, z\}$	$\{w, y, z\}$
$\{x, y, z\}$	$\{w, x, y, z\}$

The correct answer is D.

171. The number $\sqrt{63 - 36\sqrt{3}}$ can be expressed as $x + y\sqrt{3}$ for some integers x and y. What is the value of xy?

A. –18

B. –6

C. 6

D. 18

E. 27

Algebra Operations on Radical Expressions

Squaring both sides of $\sqrt{63 - 36\sqrt{3}} = x + y\sqrt{3}$ gives

$63 - 36\sqrt{3} = x^2 + 2xy\sqrt{3} + 3y^2 = (x^2 + 3y^2) + (2xy)\sqrt{3}$, which implies that $-36 = 2xy$, or $xy = -18$. Indeed, if $-36 \neq 2xy$, or equivalently, if $36 + 2xy \neq 0$, then we could write $\sqrt{3}$ as a quotient of

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$xy = -18$. Indeed, if $36 + 2xy \neq 0$, or equivalently, if $36 + 2xy \neq 0$, then we could write $\sqrt{3}$ as a quotient of the two integers $63 - x^2 - 3y^2$ and $36 + 2xy$, which is not possible because $\sqrt{3}$ is an irrational number. To be more explicit, $63 - 36\sqrt{3} = x^2 + 2xy\sqrt{3} + 3y^2$ implies $63 - x^2 - 3y^2 = (36 + 2xy)\sqrt{3}$, and if $36 + 2xy \neq 0$, then we could divide both sides of the equation $63 - x^2 - 3y^2 = (36 + 2xy)\sqrt{3}$ by $36 + 2xy$ to get $\dfrac{63 - x^2 - 3y^2}{36 + 2xy} = \sqrt{3}$.

The correct answer is A.

172. There are 10 books on a shelf, of which 4 are paperbacks and 6 are hardbacks. How many possible selections of 5 books from the shelf contain at least one paperback and at least one hardback?

A. 75

B. 120

C. 210

D. 246

E. 252

Arithmetic Elementary Combinatorics

The number of selections of 5 books containing at least one paperback and at least one hardback is equal to $T - N$, where T is the total number of selections of 5 books and N is the number of selections that do not contain both a paperback and a hardback. The value of T is

$$\begin{aligned}\binom{10}{5} &= \frac{10!}{5!\,(10-5)!} = \frac{(6)\,(7)\,(8)\,(9)\,(10)}{(1)\,(2)\,(3)\,(4)\,(5)} \\ &= (7)\,(2)\,(9)\,(2) = 252.\end{aligned}$$

To find the value of N, first note that no selection of 5 books can contain all paperbacks, since there are only 4 paperback books. Thus, the value of N is equal to the number of selections of 5 books that contain all hardbacks, which is equal to 6 since there are 6 ways that a single hardback can be left out when choosing the 5 hardback books. It follows that the number of selections of 5 books containing at least one paperback and at least one hardback is $T - N = 252 - 6 = 246$.

The correct answer is D.

173. If x is to be chosen at random from the set {1, 2, 3, 4} and y is to be chosen at random from the set {5, 6, 7}, what is the probability that xy will be even?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

E. $\frac{5}{6}$

Arithmetic; Algebra Probability; Concepts of Sets

By the principle of multiplication, since there are 4 elements in the first set and 3 elements in the second set, there are (4)(3) = 12 possible products of xy, where x is chosen from the first set and y is chosen from the second set. These products will be even EXCEPT when both x and y are odd. Since there are 2 odd numbers in the first set and 2 odd numbers in the second set, there are (2)(2) = 4 products of x and y that are odd. This means that the remaining 12 − 4 = 8 products are even. Thus, the probability that xy is even is $\frac{18}{12} = \frac{2}{3}$.

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12 3

The correct answer is D.

174. The function f is defined for each positive three-digit integer n by $f(n) = 2^x\ 3^y\ 5^z$, where x, y, and z are the hundreds, tens, and units digits of n, respectively. If m and v are three-digit positive integers such that $f(m) = 9f(v)$, then $m - v =$

A. 8

B. 9

C. 18

D. 20

E. 80

Algebra Place Value

Let the hundreds, tens, and units digits of m be A, B, and C, respectively; and let the hundreds, tens, and units digits of v be a, b, and c, respectively. From $f(m) = 9f(v)$ it follows that $2^A 3^B 5^C = 9\left(2^a 3^b 5^c\right) = 3^2\left(2^a 3^b 5^c\right) = 2^a 3^{b+2} 5^c$. Therefore, $A = a$, $B = b + 2$, and $C = c$. Now calculate $m - v$.

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